Numerical Illustration

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The file `F_simu_exact.R` contains the `R` code for the numerical evidence in Section A below. The file `permutohedron34.R` contains the `R` code to generate the permutohedron as shown in Figure 2.

A Numerical Evidence of Two Theorems

We give two simple simulations to illustrate the exact finite-sample results in Theorems A.1 and A.2 below.

We first define

$$F = \frac{\mathbf{w}^{\mathrm{T}}(\mathbf{P}_{\mathbf{Z}} - \mathbf{P}_{\mathbf{Z}_2})\mathbf{w}/q}{\mathbf{w}^{\mathrm{T}}(\mathbf{I}_n - \mathbf{P}_{\mathbf{Z}})\mathbf{w}/(n-p)} \sim F(q, n-p), \tag{A.1}$$

where we partition $\mathbf{Z}$ as $\mathbf{Z} = [\mathbf{Z}_1\ \mathbf{Z}_2]$ with $\mathbf{Z}_1 \in \mathbb{R}^{n\times q}$ and $\mathbf{Z}_2 \in \mathbb{R}^{n\times(p-q)}$, and $\mathbf{P}_{\mathbf{A}} = \mathbf{A}(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}$ denotes the projection matrix onto the column space of $\mathbf{A}$ with full column rank.

Theorem A.1. (A.1) *holds if the following two conditions hold:*

1. $\mathbf{Z}$ *is a left-spherically symmetric random matrix with* $\mathbb{P}\{\mathrm{rank}(\mathbf{Z}) = p\} = 1$*; and*
2. $\mathbf{w} \neq \mathbf{0}_n$ *is a fixed vector.*

Consider the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{A.2}$$

where $\mathbf{y} \in \mathbb{R}^n$ is the response vector, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the design matrix, $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector, and $\boldsymbol{\beta} \in \mathbb{R}^p$ is the parameter vector to be estimated. Assume $\text{rank}(\mathbf{X}) = p$. Define the ordinary least squares (OLS) estimator as

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}. \tag{A.3}$$

Given the parameter of interest $\mathbf{C}\boldsymbol{\beta}$, where $\mathbf{C} \in \mathbb{R}^{q \times p}$ has $\text{rank}(\mathbf{C}) = q \leq p$, the following F statistic is the foundation for testing hypotheses or constructing confidence regions (Ding 2026):

$$F = \frac{\{(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})^{\mathrm{T}}\mathbf{C}^{\mathrm{T}}\left[\mathbf{C}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{C}^{\mathrm{T}}\right]^{-1}\mathbf{C}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})\}/q}{[\mathbf{y}^{\mathrm{T}}\left(\mathbf{I}_n - \mathbf{P}_{\mathbf{X}}\right)\mathbf{y}]/(n-p)}. \tag{A.4}$$

Theorem A.2. *The F in (A.4) follows $F(q, n-p)$ if the following two conditinos hold:*

1. *$\mathbf{X}$ is a left-spherically symmetric random matrix with $\mathbb{P}\{\text{rank}(\mathbf{X}) = p\} = 1$; and*
2. *$\boldsymbol{\varepsilon} \neq \mathbf{0}_n$ is a fixed vector.*

Now we use two numerical examples to show confirm the results in Theorems A.1 and A.2 above. In both examples, we set $n = 8$, $p = 3$, and $q = 2$, so the theoretical reference distribution is $F(2, 5)$. We use 100,000 Monte Carlo replications. Since our theoretical results are finite-sample exact, we deliberately keep the sample size small before the asymptotics kicks in.

First, to illustrate Theorem A.1, fix $\mathbf{w} = (1, 2, \ldots, 8)^{\mathrm{T}}$. In each replication, generate $\mathbf{Z} \in \mathbb{R}^{8 \times 3}$ with i.i.d. $N(0, 1)$ entries and partition $\mathbf{Z} = [\mathbf{Z}_1 \ \mathbf{Z}_2]$, where $\mathbf{Z}_1 \in \mathbb{R}^{8 \times 2}$ and $\mathbf{Z}_2 \in \mathbb{R}^{8 \times 1}$. We then calculate the F statistic in (A.1). Theorem A.1 implies that this statistic follows exactly $F(2, 5)$.

Second, to illustrate Theorem A.2, consider the linear model (A.2) with

$$\boldsymbol{\beta} = (1, -1, 2)^{\mathrm{T}}, \qquad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \boldsymbol{\varepsilon} = (1, 2, \ldots, 8)^{\mathrm{T}}.$$

The error vector $\boldsymbol{\varepsilon}$ is fixed throughout all replications. In each replication, generate $\mathbf{X} \in \mathbb{R}^{8\times 3}$ with i.i.d. $N(0,1)$ entries, set $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, and calculate the F statistic in (A.4). Theorem A.2 implies that this statistic also follows exactly $F(2,5)$.

Figure 1 presents Q–Q plots of the simulated statistics against the $F(2,5)$ distribution. In both examples, the empirical quantiles closely follow the 45° line even with the small sample size $n = 8$, illustrating the exact finite-sample conclusions of Theorems A.1 and A.2.

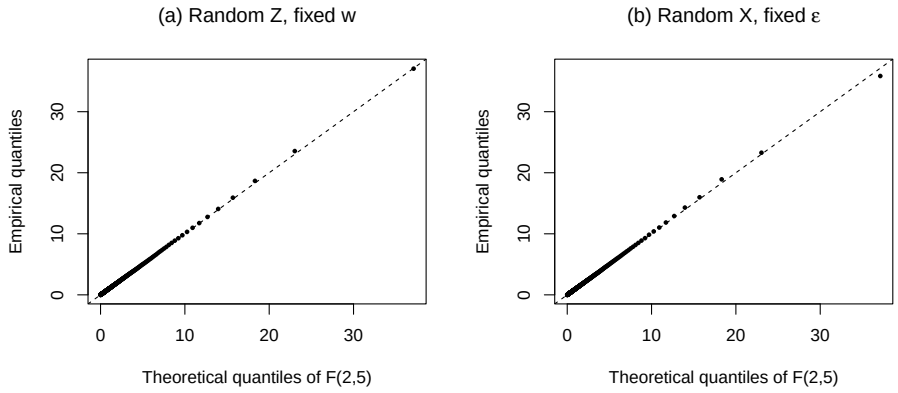


Figure 1: Q–Q plots of the simulated F statistics against the $F(2,5)$ distribution based on 100,000 Monte Carlo replications. Left: fixed $\mathbf{w}$ and random left-spherically symmetric $\mathbf{Z}$, illustrating Theorem A.1. Right: fixed $\boldsymbol{\varepsilon}$ and random left-spherically symmetric $\mathbf{X}$, illustrating Theorem A.2. The dashed line is the 45° line.

B Permutohedron Plot

References

Ding, P. (2026), *Linear Model and Extensions*, Chapman & Hall/CRC, New York. Forthcoming.

McCullagh, P. (1993), Models on spheres and models for permutations, *in* M. A. Fligner

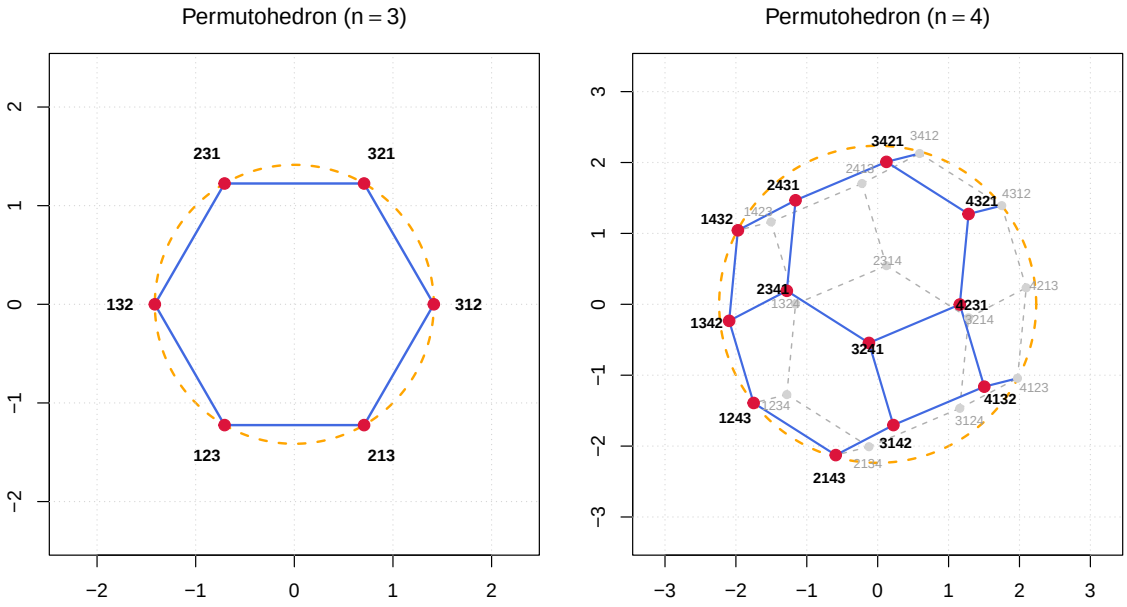


Figure 2: Geometric representations of all permutations of $(1, 2, 3)^{\mathrm{T}}$ (left) and $(1, 2, 3, 4)^{\mathrm{T}}$ (right), after centering. After rescaling, the six points for $n = 3$ lie on S^1, whereas the 24 points for $n = 4$ lie on S^2. The figure follows the geometric representation in McCullagh (1993, p. 279) and is reproduced with the help of AI.

& J. S. Verducci, eds, 'Probability Models and Statistical Analyses for Ranking Data', Vol. 80 of *Lecture Notes in Statistics*, Springer, New York, pp. 278–283.